



Miro1 Is a Molecular Marker for Parkinson's Risk.

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Summary

Background: There is a lack of reliable molecular markers for Parkinson's disease (PD) patients and at-risk individuals. The detection of the pre-symptomatic population of PD will empower more effective clinical intervention to delay or prevent disease onset. We have previously found that the mitochondrial protein Miro1 is resistant to mitochondrial depolarization-induced degradation in fibroblasts from a large number of PD patients and several at-risk individuals. Therefore, Miro1 has the potential to molecularly mark PD populations.

Objective: We aim to determine whether Miro1 is useful for labeling individuals at risk for PD.





Methods: We examine the Miro1 response to mitochondrial depolarization by biochemical approaches in induced pluripotent stem cells from a cohort of at-risk individuals.

Result: The Miro1 phenotype is significantly associated with PD risk.

Conclusion: Miro1 is a promising molecular marker for detecting both PD and at-risk populations. Tracking this Miro1 marker could aid in diagnosis and Miro1-based drug discoveries.

Method

Subject Details

No animal models or human subjects were used in this study. The iPSC work was approved by the Stanford Stem Cell Oversight Committee. iPSCs were obtained under an MTA from the National Institute of Neurological Disorders and Stroke (NINDS) human and cell repository or Parkinson's Progression Markers Initiative (PPMI), which is in a partnership with multiple institutions that deposited iPSCs, approved study protocols, and ensured consent from donors. All iPSC lines in this study are summarized in Supplemental Table 1. PPMI is an international, multi-center, and progressing study designed to identify PD biomarkers by the MJFF (<http://www.ppmi-info.org/study-design/>). The study design, subject recruitment criteria, site selection, and study assessment have been detailed in ¹.

Cell Culture and Western Blotting

iPSCs were cultured in mTeSR Plus Kit (05825, Stemcell Technologies) and maintained in a 37°C, 5% CO₂ incubator with humidified atmosphere. The media were refreshed every 1-2 days





and split every 4-6 days. CCCP (C2759, Sigma-Aldrich) was prepared at 40 mM in DMSO fresh every time and applied at 40 μ M in fresh culture medium (1:1000 dilution) for 6 hours. Cells were subsequently lysed in NP40 Cell lysis buffer (FNN0021, ThermoFisher Scientific) with protease inhibitor cocktail (539134, Calbiochem). Cell debris was removed by centrifugation at 17,000 g for 10 min at 4°C. Cell lysates were mixed 1:1 with 2 $\times$ Laemmli buffer (4% SDS, 20% Glycerol, 120 mM Tris-HCl, 0.02% bromophenol blue, 700 mM 2-mercaptoethanol) and boiled for 5 min prior to being loaded into an SDS-PAGE. 10% polyacrylamide gels (acrylamide:bis-acrylamide=29:1) and Tris-Glycine-SDS buffer (24.8 mM Tris, 192 mM glycine, 0.1% SDS) were used for electrophoresis. After electrophoresis, nitrocellulose membranes (1620115, Bio-Rad) were used in semi-dry transfer with Bjerrum Schafer-Nielsen buffer (48 mM Tris, 39 mM glycine, 20% Methanol (v/v), pH 9.2). Transferred membranes were first blocked overnight in phosphate-buffered saline (PBS) containing 5% fat-free milk and 0.1% tween-20 at 4°C, and then incubated with the following primary antibodies: mouse anti-Miro1 (WH0055288M1, Sigma-Aldrich) at 1:1,000, or rabbit anti-GAPDH (5174S, Cell Signaling Technology) at 1:1-3,000, at 4°C overnight in blocking buffer. HRP-conjugated goat anti-mouse (115-035-003, Jackson ImmunoResearch) or goat anti-rabbit (111-035-144, Jackson ImmunoResearch Laboratories) were used at 1:10-20,000. Pierce ECL Western Blotting Substrate (32109, ThermoScientific) were used for ECL immunoblotting. Membranes were scanned using a Bio-Rad ChemiDoc XRS system. Experiments were repeated for more than 3 times. Cell passaging numbers were within the range of 12-17 which had no influence on the phenotype.





Quantification of Western Blotting Data

All experiments were performed in a blinded format. The intensities of protein bands were measured by ImageJ (ver. 1.48V, NIH). The intensity of each band was normalized to that of the loading control GAPDH from the same blot, and expressed as a fraction of that of “Healthy-1 with DMSO treatment” from the same experiment; this control was included in every independent experiment. Values of Mean $\pm$ S.E.M of Miro1 were reported in Supplemental Table 1. The ratio of Miro1 was calculated by dividing the mean of Miro1 intensities treated with CCCP by the mean of Miro1 with DMSO of the same subject, and imported into the heat map. Student T Test was performed for comparing normalized Miro1 band intensities within the same subject (“with DMSO” v.s. “with CCCP”), and P values were reported in Supplemental Table 1. The number of subjects with a P value >0.05 together with subjects that showed significant Miro1 upregulation after CCCP, or the number of subjects with $P<0.05$ for Miro1 reduction after CCCP was counted respectively and used in Fisher Exact Test. n=3-55 independent experiments.

Multivariate regression or Anova was used to determine the interactions among multiple variables for affecting Miro1 ratio and P values were calculated by linear fit in Fig. 2-3. During the analysis, Hoehn and Yahr Scale and Mini-Mental Status Examination were detected to show an interaction. Partial regression plots were subsequently generated to help decipher the relationship between an individual variable and the response variable in a multivariable regression problem. Seven partial regression plots, one for each individual variable in the regression problem (Hoehn and Yahr Scale, Mini-Mental Status Examination, onset age), their interaction terms, and the intercept were generated. Out of all the partial regression plots for





interaction terms, the plot of Hoehn and Yahr Scale and Mini-Mental Status Examination with linear fit showed significance ($P=0.012$).

Enzyme-Linked Immunosorbent Assay (ELISA)

All experiments were performed as blinded tests. 40 μ M CCCP in DMSO or the same volume of DMSO alone was applied to iPSCs for 6 hours, and then cells were lysed in NP40 Cell lysis buffer (FNN0021, ThermoFisher Scientific) with protease inhibitor cocktail (539134, Calbiochem). Cell debris was removed by centrifugation at 17,000 g for 10 min at 4°C. The Rhot1 ELISA kit (EKL54911, Biomatik) was used according to the manufacturer's instructions. The specificity and stability were validated by Biomatik. The dynamic detection range, sensitivity (lower limit of detection–LLOD), and precision (inter- and intra-assay) were determined by both Biomatik and us, and the results were comparable. Briefly, 50 μ l of cell lysate prepared from above, or serial dilutions of the standard (0–40 ng/ml) were added and incubated for 2 hours at 37°C. Each well was then incubated with 100 μ l of Detection Reagent A for 1 hour at 37°C. Next, plates were washed, and each well was incubated with 100 μ l of Detection Reagent B for 1 hour at 37°C. Plates were washed again, and 90 μ l of Substrate Solution was added to each well for 15–25 minutes at 37°C. The colorimetric reactions were stopped by 50 μ l of Stop Solution and absorbance was read at 450 nm by a microplate reader (Infinite 200 Pro, Tecan). An experiment for generating the standard curve was included in each plate and the representative standard plot was shown in the figure. Each data point was from 3–4 independent experiments with 2 technical repeats each time. Student T Test was performed for comparing Miro1 signals within the same subject (DMSO vs. CCCP). A loading control,





GAPDH, was detected by Western blotting in each experiment and there was no significant difference in loading between “DMSO” and “CCCP” for each cell line. Raw data and P values are reported in Supplemental Table 1.

References

1. Parkinson Progression Marker, I. The Parkinson Progression Marker Initiative (PPMI). *Progress in neurobiology* **95**, 629-635 (2011).

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